MTH 336(A) Dr. James Ashe
Worksheet 12.3 Monday 11$^{\text{th}}$ November, 2013 **Name:**

Limits of functions *Evaluate the following limits.*

1. $\lim_{(x,y)\to(1,-3)} (3x + 4y - 2)$

Limits at boundary points *Evaluate the following limits.*

2. $\lim_{(x,y)\to(2,2)} \dfrac{y^2 - 4}{xy - 2x}$

3. $\lim_{(x,y)\to(4,5)} \dfrac{\sqrt{x+y} - 3}{x + y - 9}$

Nonexistence of limits *Use the Two path test to prove that the following limits do not exist.*

4. $\lim_{(x,y)\to(0,0)} \dfrac{x + 2y}{x - 2y}$

5. $\lim_{(x,y)\to(0,0)} \dfrac{4xy}{3x^2 + y^2}$

6. $\lim_{(x,y)\to(0,0)} \dfrac{y}{\sqrt{x^2 - y^2}}$